

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical
M2 Practical-8 [Mini Project]

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Class: TYIT	Batch: 1
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Project Title: College Canteen Pre-Order System

Develop a web-based College Canteen Pre-Order System. It should allow students to:

1. View the available food items and their prices.
2. Enter student details such as name, roll number, and course.
3. Select a food item and quantity.
4. Place an order using a POST API.
5. Store the order details in MongoDB.
6. Retrieve the stored orders using a GET API.
7. Display the available food items and placed orders on the React page.

APIs to be implemented

- GET /food – Retrieve food items.
- POST /orders – Add a new food order.
- GET /orders – Retrieve placed orders.

Technologies

- React
- Axios
- Node.js
- Express.js
- MongoDB

Expected Output

The final application should contain:

- Food Item List
- Student Order Form
- Place Order button
- Order List/Table

Design the interface and decide the appropriate fields, MongoDB document structure, and API flow.

Code:

server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
```

```
app.use(cors());
app.use(express.json());

mongoose.connect("mongodb://127.0.0.1:27017/college")
  .then(() => console.log("MongoDB Connected"));

const Order = mongoose.model("students", new mongoose.Schema({
  studentName: String,
  rollNo: Number,
  course: String,
  foodItem: String,
  price: Number,
  quantity: Number,
  total: Number
}));

const food = [
  { name: "Pizza", price: 100 },
  { name: "Veg Burger", price: 60 },
  { name: "Sandwich", price: 50 },
  { name: "Samosa", price: 20 }
];

// GET /food
app.get("/food", (req, res) => {
  res.json(food);
});

// POST /orders
app.post("/orders", async (req, res) => {
  const { studentName, rollNo, course, foodItem, quantity } = req.body;

  const item = food.find(f => f.name === foodItem);
  const total = item.price * quantity;

  const order = new Order({
    studentName,
    rollNo,
    course,
    foodItem,
    price: item.price,
    quantity,
    total
  });
```

```
    await order.save();
    res.json(order);
  });

// GET /orders
app.get("/orders", async (req, res) => {
  const orders = await Order.find();
  res.json(orders);
});

app.listen(5000, () => console.log("Server running on 5000"));
```

App.jsx:-

```
import { useEffect, useState } from "react";
import axios from "axios";
import "./App.css";

function App() {
  const [food, setFood] = useState([]);
  const [orders, setOrders] = useState([]);

  const [form, setForm] = useState({
    studentName: "",
    rollNo: "",
    course: "",
    foodItem: "",
    quantity: 1
  });

  useEffect(() => {
    axios.get("http://localhost:5000/food")
      .then(res => setFood(res.data));

    getOrders();
  }, []);

  const getOrders = () => {
    axios.get("http://localhost:5000/orders")
      .then(res => setOrders(res.data));
  };

  const change = e => {
    setForm({ ...form, [e.target.name]: e.target.value });
  };
}
```

```
};

const order = e => {
  e.preventDefault();

  axios.post("http://localhost:5000/orders", {
    ...form,
    rollNo: Number(form.rollNo),
    quantity: Number(form.quantity)
  }).then(() => {
    alert("Order Placed!");
    getOrders();

    setForm({
      studentName: "",
      rollNo: "",
      course: "",
      foodItem: "",
      quantity: 1
    });
  });
};

return (
  <div className="container">

    <h1>College Canteen Pre-Order System</h1>

    <h2>Food Items</h2>

    <div className="food">
      {food.map(f => (
        <div>
          <b>{f.name}</b>
          <p>₹ {f.price}</p>
        </div>
      ))}
    </div>

    <h2>Student Order Form</h2>

    <form onSubmit={order}>
      <input
        name="studentName"
```

```
placeholder="Student Name"
value={form.studentName}
onChange={change}
required
/>

<input
  name="rollNo"
  placeholder="Roll No"
  value={form.rollNo}
  onChange={change}
  required
/>

<input
  name="course"
  placeholder="Course"
  value={form.course}
  onChange={change}
  required
/>

<select
  name="foodItem"
  value={form.foodItem}
  onChange={change}
  required
>
  <option value="">Select Food</option>
  {food.map(f => (
    <option value={f.name}>
      {f.name} - ₹ {f.price}
    </option>
  ))}
</select>

<input
  type="number"
  name="quantity"
  min="1"
  value={form.quantity}
  onChange={change}
/>
```

```
    <button>Place Order</button>
  </form>

  <h2>Placed Orders</h2>

  <table>
    <thead>
      <tr>
        <th>Name</th>
        <th>Roll No</th>
        <th>Course</th>
        <th>Food</th>
        <th>Price</th>
        <th>Qty</th>
        <th>Total</th>
      </tr>
    </thead>

    <tbody>
      {orders.map(o => (
        <tr>
          <td>{o.studentName}</td>
          <td>{o.rollNo}</td>
          <td>{o.course}</td>
          <td>{o.foodItem}</td>
          <td>₹{o.price}</td>
          <td>{o.quantity}</td>
          <td>₹{o.total}</td>
        </tr>
      ))}
    </tbody>
  </table>

</div>
);
}
```

```
export default App;
```

App.css:

```
body {
  font-family: Arial;
  background: #f2f2f2;
}
```

```
.container {  
  width: 90%;  
  margin: auto;  
}
```

```
h1 {  
  text-align: center;  
  color: #333;  
}
```

```
.food {  
  display: flex;  
  gap: 15px;  
}
```

```
.food div {  
  background: white;  
  padding: 15px;  
  width: 150px;  
  border-radius: 8px;  
}
```

```
form {  
  background: white;  
  padding: 20px;  
  margin: 20px 0;  
}
```

```
input, select, button {  
  display: block;  
  width: 100%;  
  padding: 10px;  
  margin: 8px 0;  
  box-sizing: border-box;  
}
```

```
button {  
  background: green;  
  color: white;  
  border: none;  
  cursor: pointer;  
}
```

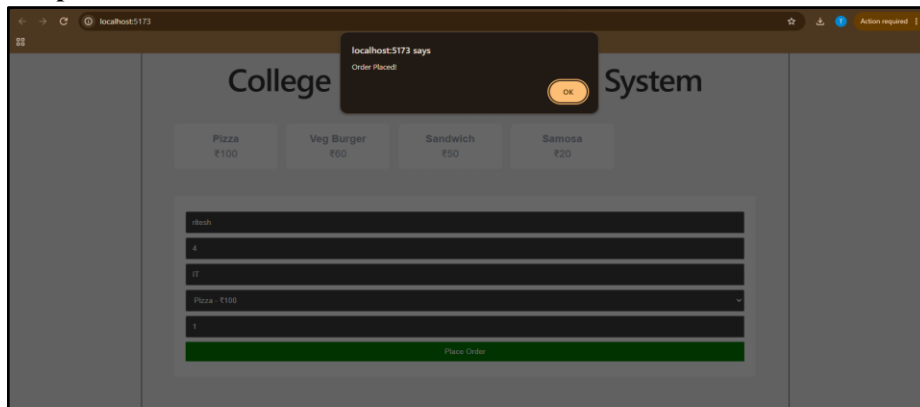
Asmitha T002

```
table {  
  width: 100%;  
  background: white;  
  border-collapse: collapse;  
}
```

```
th, td {  
  padding: 10px;  
  border: 1px solid #ccc;  
  text-align: center;  
}
```

```
th {  
  background: #333;  
  color: white;  
}
```

Output:



Select Food						
1						
Place Order						
Name	Roll No	Course	Food	Price	Qty	
Asmitha	13	B.Sc. IT	Pizza	₹100	2	
Radhika	5	B.Sc. IT	Veg Burger	₹60	1	
Afsha	16	B.Sc. IT	Sandwich	₹50	3	
Aashu	14	B.Sc. IT	Samosa	₹20	4	